GE VERNOVA

ADMS

16.25.1

Group Control User Guide

Copyright

Change History

Date	Change Description
Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Mar 25, 2024	ADMS 16.7.0: • Updated document format and styling.

Contents

1. Introduction

The Group Control application provides the ability to issue a group control for SCADA devices. This allows an operator to work more efficiently by being able to issue group controls and perform similar operations on a group of devices by circuit (feeder), power transformer, substation, or area of responsibility.

The Group Control display includes a hierarchical tree with stations, transformers, and feeders where an operator can filter devices by device type or an area and create a list of devices to be controlled as a group. When issuing a group control, the DMS server uses the ISD link to forward the control to Scada, which then controls the devices using real-time units (RTUs).

For example, operators can use the Group Control display to do the following:

- Disable switching devices participating in automated switching schemes and teams before load shed.
- Individually issue a device poll request (demand scan) for one device at a time from the filtered results.
- Enable all reclosers downstream of the low side of a transformer.
- Open all switches downstream of a feeder.

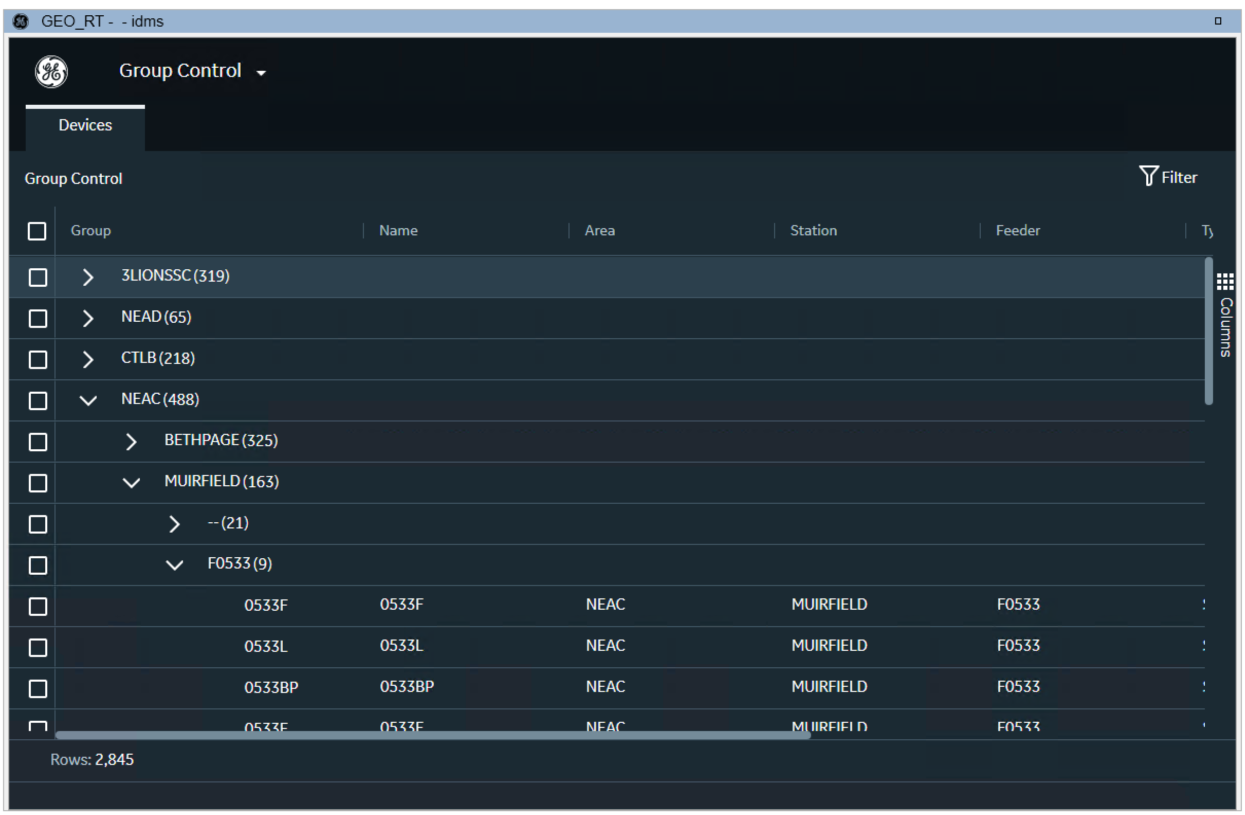


Figure 1. Group Control Application

2. Setting Up the Group Control Application

> **Important**
>
> The Group Control application (sometimes referred to as GC) runs in scope of the ADMS system and controls devices through the Scada system. Therefore, these setup instructions assume that you already have the ADMS and Scada systems up and running.

The Group Control application user interface is an Angular web application hosted on the Apache httpd server.

The following sections provide details for setting up Group Control.

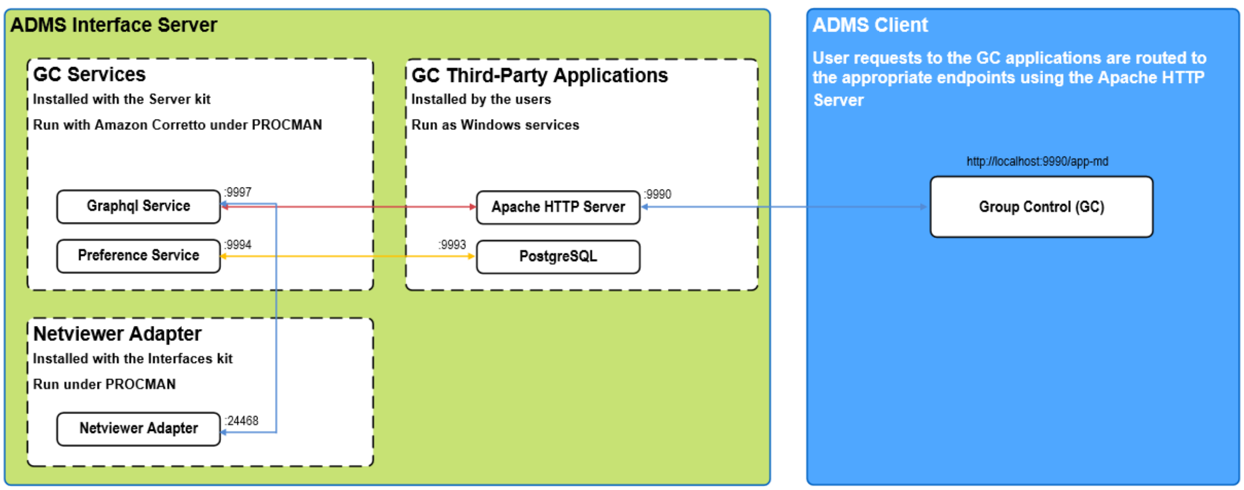


Figure 2. Group Control Flow

Apache HTTP web server is used for the web server hosting and for proxying requests from Net Clients (ADMS clients) to ADMS backend services. All clients must have access to the host and port of the Apache HTTP web server. The HTTPS certificate used by Apache HTTP web server must be valid for all clients and match the host name used in the CnvHostName parameter in the netconf.xml file.

The Preference service stores and retrieves preference data from the PostgreSQL database.

Netviewer Adapter connects to the DMS server and provides DMS data to the Graphql service.

The Graphql service connects to Netviewer Adapter and communicates DMS data in GraphQL format to the Group Control application.

2.1. Installing Prerequisites

The following components are required for setting up and running the Group Control application:

- **ADMS Interfaces kit:** Installs the Netviewer Adapter application and Group Control customization files.

- **Amazon Corretto:** Runs the Preference service and Graphql service Java executable files.
- **Apache HTTP Server:** Directs user requests in APF to the appropriate endpoint (for example, the Graphql service).
- **PostgreSQL:** Stores the preference data.

2.1.1. Installing the Interfaces Kit

Install the Interfaces kit with the default installation options.

The ADMS_Interfaces.msi kit includes the following components that are required for setting up and running the Group Control application:

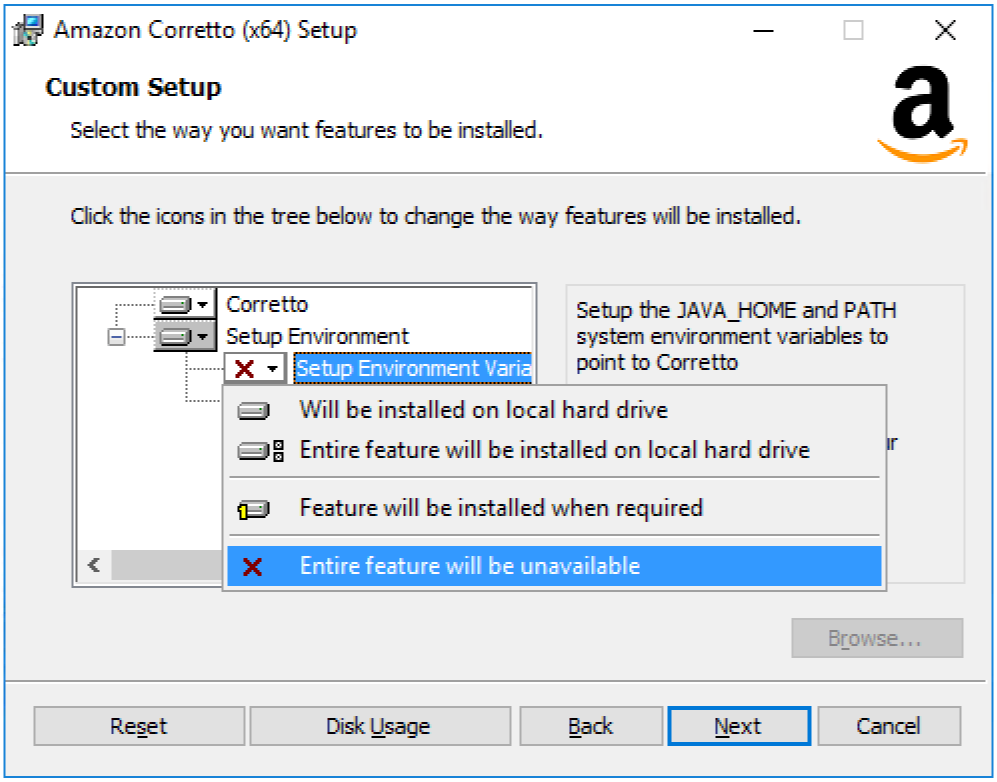
- **Netviewer Adapter:** This is the primary interface between the DMS server and the Graphql application to provide DMS data to the Group Control application.
- **Group Control Customization Files:** Files with the Group Control application settings are installed in the %IDMS_ROOT%\config\interfaces\StormAssist folder.

2.1.2. Installing Amazon Corretto

Amazon Corretto is required to run the Preference service and Graphql service Java executable files.

To install Amazon Corretto:

1. Download amazon-corretto-11.0.8.10.1-windows-x64.msi from https://github.com/corretto/corretto-11/releases.
2. Run the installer.
3. On the Custom Setup parameters page, choose the "Entire feature will be unavailable" option for the "Setup Environment Variables" feature.



4. Update the installation path so that the installation folder path and name do not include spaces (for example, install to C:\Corretto).
5. Install the application.
6. Add the CORRETTO_HOME user environment variable with the path `<Corretto_Installation_Folder>\jdk11.0.8_10\bin`. For example:

```
setx CORRETTO_HOME c:\Corretto\jdk11.0.8_10\bin
```

> **Important**
>
> PROCMAN cannot resolve CORRETTO_HOME paths with spaces.

2.1.3. Installing PostgreSQL

> **Important**
>
> GC requires a separate PostgreSQL instance to be running. It does not support using the same PostgreSQL instance as another application, such as Internal Notifications.

To install PostgreSQL:

1. Download a PostgreSQL installer from https://www.postgresql.org/download/windows/.
2. Install PostgreSQL to a local folder (for example,

D:\Eterra\Distribution\GroupControlPrerequisites\Postgres\bin).

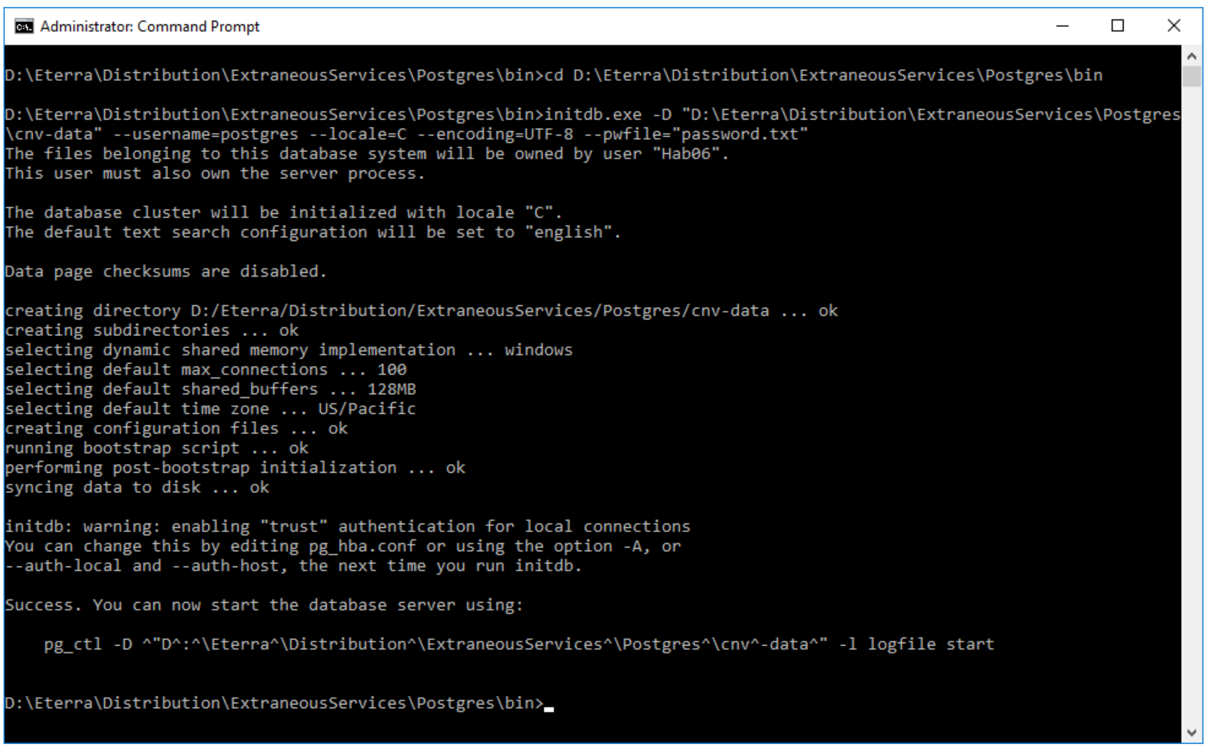
To initialize the PostgreSQL database:

1. Open a command prompt as Administrator.
2. Navigate to the PostgreSQL bin folder (for example, D:\Eterra\Distribution\GroupControlPrerequisites\Postgres\bin).
3. Create the password.txt file with an entry for the superuser database password.

   > **Note**
   >
   > The password value must match the password value in %IDMS_ROOT%\misc\CNV\procmanPref.

4. Run the following command:

   ```
   initdb.exe -D "D:\Eterra\Distribution\GroupControlPrerequisites\Postgres\cnv-data"
   --username=postgres --locale=C --encoding=UTF-8 --pwfile="password.txt"
   ```

5. Edit the postgresql.conf file in the D:\Eterra\Distribution\GroupControlPrerequisites\Postgres\cnv-data\ folder:

   - Uncomment the "port" parameter and set the port value to 9993 to match the port value in the %IDMS_ROOT%\misc\CNV\procmanPref file. For example:

     ```
     port = 9993
     ```

This completes the database initialization.

To install the PostgreSQL service:

1. Open a command prompt as Administrator.
2. Navigate to the PostgreSQL bin folder (for example, D:\Eterra\Distribution\GroupControlPrerequisites\Postgres\bin).
3. To install the service, run one of the following commands:
   - To run the service automatically:

     ```
     pg_ctl.exe register -N "CNV-PostgreSQL" -S a -D
     "D:\Eterra\Distribution\GroupControlPrerequisites\Postgres\cnv-data" -w
     ```

   - To install the service as AUTHORITY\NetworkService user:

     ```
     pg_ctl.exe register -N "CNV-PostgreSQL" -U "NT AUTHORITY\NetworkService" -D
     "D:\Eterra\Distribution\GroupControlPrerequisites\Postgres\cnv-data" -w
     ```

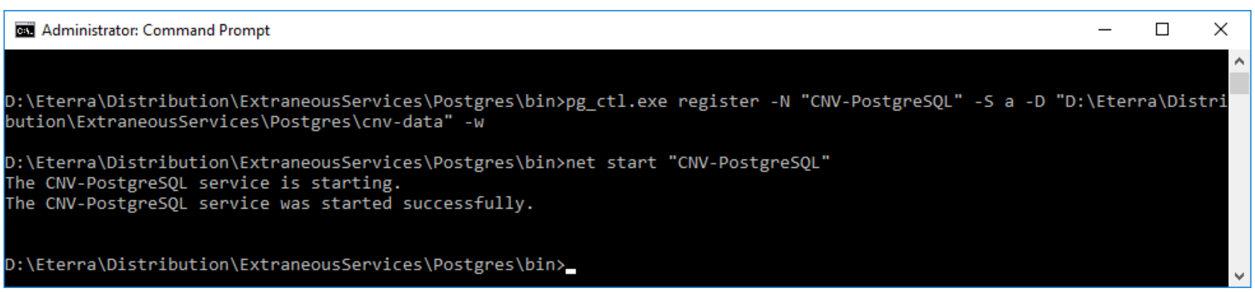
   - To run the service manually, on demand:

     ```
     pg_ctl.exe register -N "CNV-PostgreSQL" -S d -D
     "D:\Eterra\Distribution\GroupControlPrerequisites\Postgres\cnv-data" -w
     ```

4. Start the service:

   ```
   net start "CNV-PostgreSQL"
   ```

   Alternatively, you can start the PostgreSQL service from the Windows Services display.



To uninstall the PostgreSQL service:

1. Open a command prompt as Administrator.
2. Run the following command:

   ```
   sc delete "CNV-PostgreSQL"
   ```

   Alternatively, you can uninstall the service using pg_ctl, by running the following command from the PostgreSQL bin folder:

```
pg_ctl.exe unregister -N "CNV-PostgreSQL"
```

2.1.4. Installing Apache HTTP Server

For the default configuration, https://localhost:9990 represents the Apache HTTP web server hosting address. In a distributed environment, the Apache HTTP web server can be hosted on a separate host. The configuration also contains proxy configurations required for the Group Control web application.

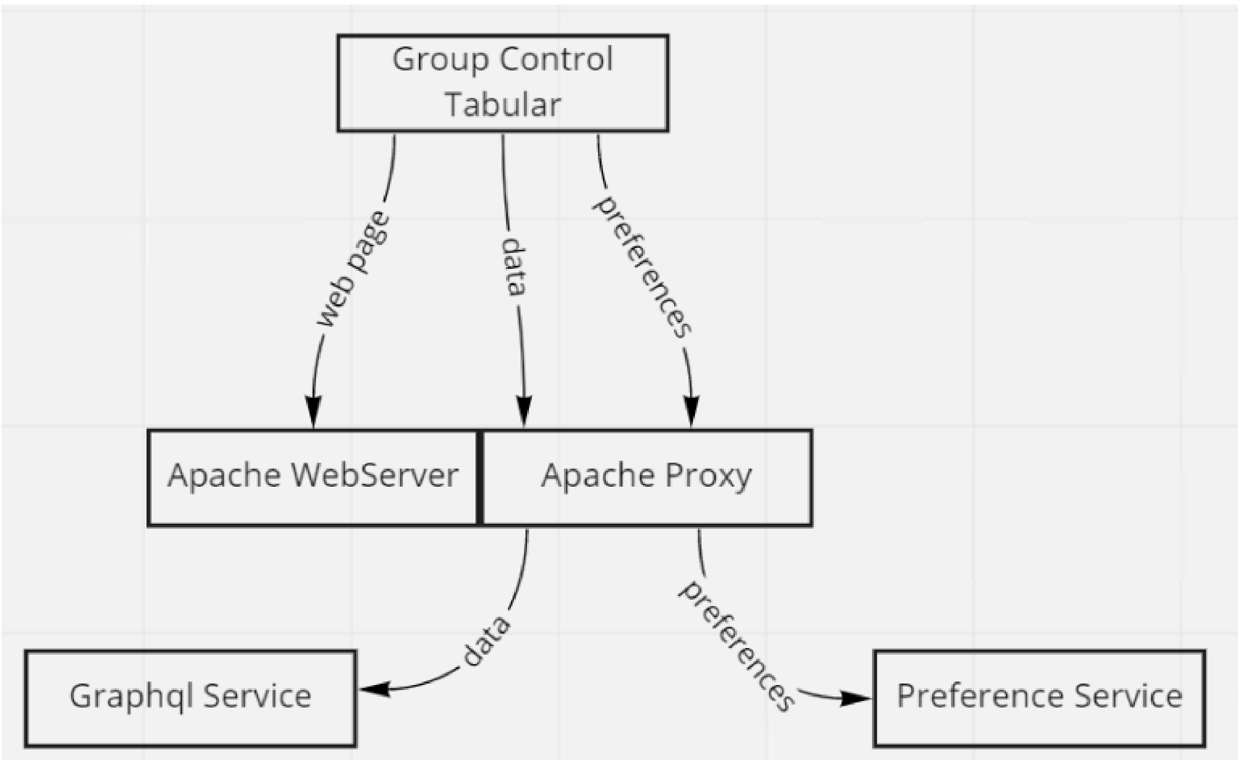


Figure 3. Apache HTTP Server Flow

To install the Apache HTTP Server:

1. Download the Apache HTTP Server installer from https://httpd.apache.org/download.cgi. For example, download httpd-2.4.52-o111m-x64-vc15.zip from https://www.apachehaus.com/cgi-bin/download.plx, which is an official mirror of Apache builds for Windows.
2. Install the Apache HTTP Server to a local folder (for example, D:/Eterra/Distribution /GroupControlPrerequisites /Apache).
3. Add the SRVROOT user environment variable with the path "<Apache HTTP Server_Installation_Folder>". For example:

```
setx SRVROOT D:/Eterra/Distribution/GroupControlPrerequisites/Apache
```

To configure the Apache HTTP Server:

Do one of the following:

- Overwrite default httpd.conf file in D:\Eterra\Distribution\GroupControlPrerequisites\Apache\conf with the demo httpd.conf file. For the demo httpd.conf file, see Demo httpd.conf Configuration.
- Edit the default httpd.conf file in D:\Eterra\Distribution\GroupControlPrerequisites\Apache\conf:
  - Set the listening port to 9990 and enable the HTTPS mode:

    ```
    Listen 9990 https
    ```

> **Note**
>
> You do not need to modify the ServerRoot parameter, which points to the SRVROOT environment variable.

  - Set HTTPS (SSL) settings as follows (To run in HTTPS mode, the Group Control requires certificates to be referenced in the configuration file and recorded in the system certificate store).

    ```
    <VirtualHost _default_:9990>
        ServerName localhost
        SSLEngine on
        SSLCertificateFile "${SRVROOT}/conf/adms-cnv-server.crt"
        SSLCertificateKeyFile "${SRVROOT}/conf/adms-cnv-server.key"
        SSLCertificateChainFile "${SRVROOT}/conf/adms-root-ca.crt"
        # enable HTTP/2, if available
        Protocols h2 http/1.1
        # HTTP Strict Transport Security (mod_headers is required) (63072000 seconds)
        Header always set Strict-Transport-Security "max-age=63072000"
    </VirtualHost>

    SSLProtocol             all -SSLv3 -TLSv1 -TLSv1.1
    SSLHonorCipherOrder     off
    SSLSessionTickets       off
    SSLUseStapling On
    SSLStaplingCache "shmcb:logs/ssl_stapling(32768)"
    ```

  - Set the configuration entries as follows:

    ```
    DocumentRoot "$\{IDMS_ROOT}/webpages"
    <Directory "$\{IDMS_ROOT}/webpages">
    ```

  - Add the following entries at the bottom of the configuration file:

    ```
    ProxyPass "/preferences" "http://localhost:9994/preferences"
    ProxyPassReverse "/preferences" "http://localhost:9994/preferences"
    ProxyPass "/v2/graphql" "http://localhost:9997/v2/graphql"
    ProxyPassReverse "/v2/graphql" "http://localhost:9997/v2/graphql"
    ```

- Optionally, to run GC in HTTP mode:
    - Remove "https" from after the listening port:

      ```
      Listen 9990
      ```

    - Comment the HTTPS (SSL) configurations:

      ```
      # <VirtualHost _default_:9990>
      ...
      # SSLStaplingCache "shmcb:logs/ssl_stapling(32768)"
      ```

- Optionally, configure load balancing for services using the Proxy Balancer entries.

To install the Apache HTTP Server as a service:

1. Open a command prompt as Administrator.
2. Navigate to the Apache HTTP Server bin folder (for example, D:\Eterra\Distribution\GroupControlPrerequisites\Apache\bin).
3. Run the following command:

   ```
   httpd.exe -k install -n "CNV-Apache-Httpd" -f
   "D:\Eterra\Distribution\GroupControlPrerequisites\Apache\conf\httpd.conf"
   ```

To start the Apache HTTP Server service:

- Run the following command from the Apache HTTP Server bin folder:

  ```
  httpd.exe -k start -n "CNV-Apache-Httpd"
  ```

To restart the Apache HTTP Server service:

- Run the following command from the Apache HTTP Server bin folder:

  ```
  httpd.exe -k restart -n "CNV-Apache-Httpd"
  ```

To stop the Apache HTTP Server service:

- Run the following command from the Apache HTTP Server bin folder:

  ```
  httpd.exe -k stop -n "CNV-Apache-Httpd"
  ```

To uninstall the Apache HTTP Server service:

- Run the following command from the Apache HTTP Server bin folder:

  ```
  httpd.exe -k uninstall -n "CNV-Apache-Httpd"
  ```

2.2. Configuring the System

2.2.1. Configuring GC Services

The GC services (Preference Service and Graphql Service) are installed as Java executable files (preferences-service.jar and tabular-graphql-server.jar) in the server\bin folder with the Server kit (the CNV Services option must be enabled when you install the kit).

GC service configuration files for PROCMAN are installed in %IDMS_ROOT%\misc\CNV.

The installation also creates the following environment variables, to run the GC services using PROCMAN (with the default value set to FALSE):

```
PROCMAN_FLAG_EMS_RUN_CNVPREF
PROCMAN_FLAG_EMS_RUN_CNVGRAPH
```

To automatically start GC services using PROCMAN:

- Run the following commands:

```
setx PROCMAN_FLAG_EMS_RUN_CNVPREF TRUE
setx PROCMAN_FLAG_EMS_RUN_CNVGRAPH TRUE
```

2.2.2. Customizing GC

To run GC, you must record GC preferences in the PostgreSQL database. Before recording, the preferences must be exported using the OMS Client Configuration Editor application. Configuration files with default preferences are stored in %IDMS_ROOT%\config\interfaces\StormAssist. You can adjust the preferences, as appropriate.

The following table lists GC preferences files that can be used for customization.

File	Description
GroupControlTableConfig.json	Configuration for the group device control table.
GroupControlMenuHandlerConfig.json	Configuration for the group device control table context menu.

To export GC preference files:

> **Important**
>
> Before exporting preferences, verify that sample preferences are present in the %IDMS_ROOT%\config\interfaces\StormAssist folder (installed by the Interfaces kit).

1. Open the OMS Client Configuration Editor.
2. From the File menu, select Export to JSON Data.

GC customization files with default and project-specific information (applies to coded translation items) are saved to the %IDMS_ROOT%\config\output folder.

3. Adjust the preferences, as appropriate.

To record GC preferences in the PostgreSQL database:

1. Run Windows PowerShell.
2. Navigate to %IDMS_ROOT%\misc\CNV.
3. Run the following command:

```
.\LoadPreferences.ps1
```

2.2.3. Configuring the DMS Server

To configure the DMS Server (NetSrv):

- Edit the NetServerMainConfig.txt file in %IDMS_ROOT%\config\server\:
  - Use the ISD_CTRL_STATUS_ON_OFF parameter to set the list of comma-separated mappings between the device On/Off status measurements and device On/Off status names.

    The mapping format is:

    <MeasTypeEnumeration>:<OnDisplayNameString>#<OnInternalNameString>:<OffDisplayNameString>#<OffInternalNameString>

    For example, the mapping value for the SwitchStatusMeasurementType measurement is:

    ```
    12:Closed#CLOSE:Open#TRIP
    ```

    The default value is:

    ```
    ISD_CTRL_STATUS_ON_OFF  12:Closed#CLOSE:Open#TRIP,14:On#CPON:Off#CPOF,
    111:Enabled#ENABLE:Disabled#DISABLE
    ```

  - Use the MEASUREMENT_INVERT_PNTNAM parameter to set the list of comma-separated measurement point types whose Scada values need to be inverted in ADMS. This parameter value must match the parameter value in NetServerOmsConfig.txt.

    For example, to invert measurement values for the point types ARE and GTE:

    MEASUREMENT_INVERT_PNTNAM ARE,GTE

2.2.4. Configuring the OMS Server

To configure the OMS Server (NetServer):

-

Edit the NetServerOmsConfig.txt file in %IDMS_ROOT%\config\server\:

- Use the MEASUREMENT_INVERT_PNTNAM parameter to set the list of comma-separated measurement point types whose Scada values need to be inverted in ADMS. This parameter value must match the parameter value in NetServerMainConfig.txt.

  For example, to invert measurement values for the point types ARE and GTE:

  ```
  MEASUREMENT_INVERT_PNTNAM  ARE,GTE
  ```

2.2.5. Configuring Netviewer Adapter

Netviewer Adapter provides DMS data to the Graphql service. For this purpose, Netviewer Adapter does not require tile capabilities; therefore, tile features should be disabled for performance reasons.

To configure the Netviewer Adapter:

- Edit the NetviewerAdapterConfig.xml file in %IDMS_ROOT%\config\interfaces\:
  - Specify the DMS Interface Server host and port:

    ```
    <DmsInterfaceServerUrl>http://localhost:8801</DmsInterfaceServerUrl>
    ```

  - Disable Netviewer Adapter from running in HTTPS mode:

    ```
    <SecureWS>false</SecureWS>
    ```

  - Disable Common Notification Framework and tile features:

    ```
    <DisableCnf>true</DisableCnf>
    <EnablePublishingToTileService>false</EnablePublishingToTileService>
    ```

2.2.6. Configuring the ADMS Client

To configure the ADMS client:

- Edit the netconf.xml file in %IDMS_ROOT%\config\client\:
  - Configure the GC host, port, and protocol value. For example:

    ```
    <CnvHostName>localhost</CnvHostName>
    <CnvHostPort>9990</CnvHostPort>
    <CnvHostProtocol>https</CnvHostProtocol>
    ```

    The ADMS client uses the GC host, port, and protocol values for the %webserver alias in GC Netview commands.

2.2.7. Configuring the Browser Toolbar

To configure the Browser toolbar:

- Edit the emp_toolbar.xml file in D:\Eterra\habweb06\config\toolbars\emp_toolbar.xml\:
  - Make sure that the Group Control entry exists in the OMS menu. For example:

    ```
    <MENU>
    <Command>DISPLAY/VIEWPORT=GEO_RT/POPUP /URL "NETVIEW://RT/WEB?URL=%webserver/app-md/#/group-control"</Command>
    <menuItem>Group Control Tabular</menuItem>
    </MENU>
    ```

    When a user selects this option, the Group Control display opens as a pop-up.

  - Alternatively, to open the Group Control application inside of an APF viewport, modify the Netview command as follows:

    ```
    <Command>NETVIEW://RT/WEB?URL=%webserver/app-md/#/group-control</Command>
    ```

    > **Note**
    >
    > By default, the %webserver alias in the GC Netview commands resolves to https://localhost:9990.

2.3. TCP/IP Default Port List

This section lists the default ports used by the Group Control application. These ports are listed in the Group Control configuration sections earlier in this document.

> **Note**
>
> It is the responsibility of the utility to whitelist all incoming connections to the listening TCP/IP ports by ADMS applications.

> **Important**
>
> Applications and endpoints need to be configured to use unique port values.

- ADMS interface server machine (host for GC):
  - Apache HTTP Server: 9990
  - PostgreSQL: 9993
  - Preference service: 9994
  - Graphql service: 9997
  -

Netviewer Adapter: 24468

Note

To clearly distinguish GC-specific ports from other ports, GC ports start with "999".

3. Using the Group Control Application

3.1. Running the Group Control Application

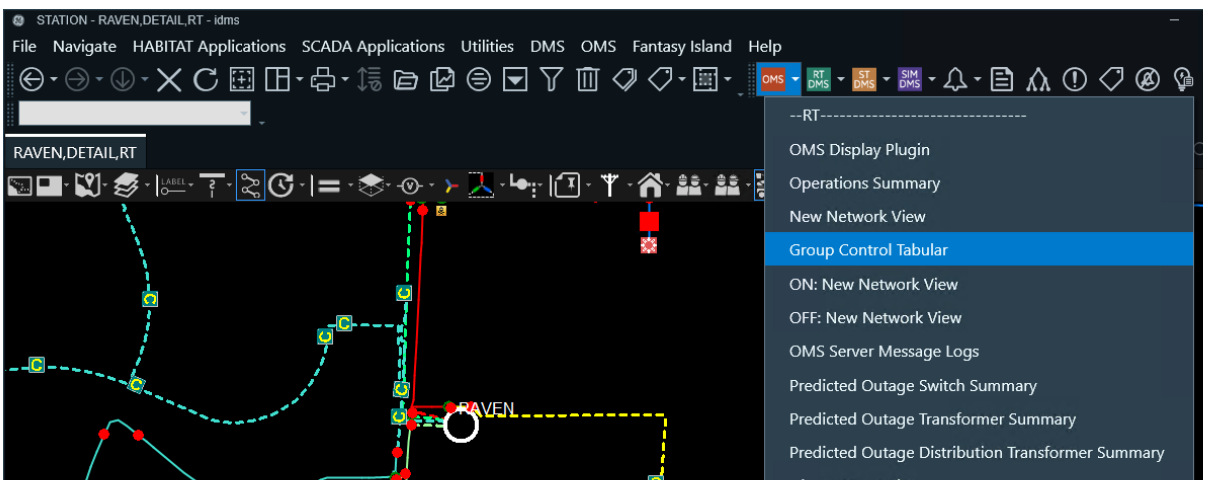
1. Run the PostgreSQL and Apache HTTP Server services.
2. Run the ADMS servers, including the NetSrv (DMS Server).
3. Run the ADMS client.
4. Run the DMS Interface Server.
5. Run the Netviewer Adapter application.
6. Run the GC services (Preference Service and Graphql Service).

> **Note**
>
> Amazon Corretto must be installed to run GC services.

3.2. Calling the Group Control Display

To show the Group Control display:

- From the Browser toolbar, click the OMS menu button, and select Group Control Tabular.



This opens the Group Control display in a new pop-up.

> **Note**
>
> The Group Control application may take several minutes to load.

3.3. Viewing Data in the Group Control Display

The Group Control display shows data for all SCADA devices that can be controlled using the Group Control application.

The Group Control display shows the data in a hierarchical manner. The hierarchy tree includes service centers, stations, feeders, and components.

You can expand any node level to see the details. For example, you can expand the feeder node to see all the devices under the feeder:

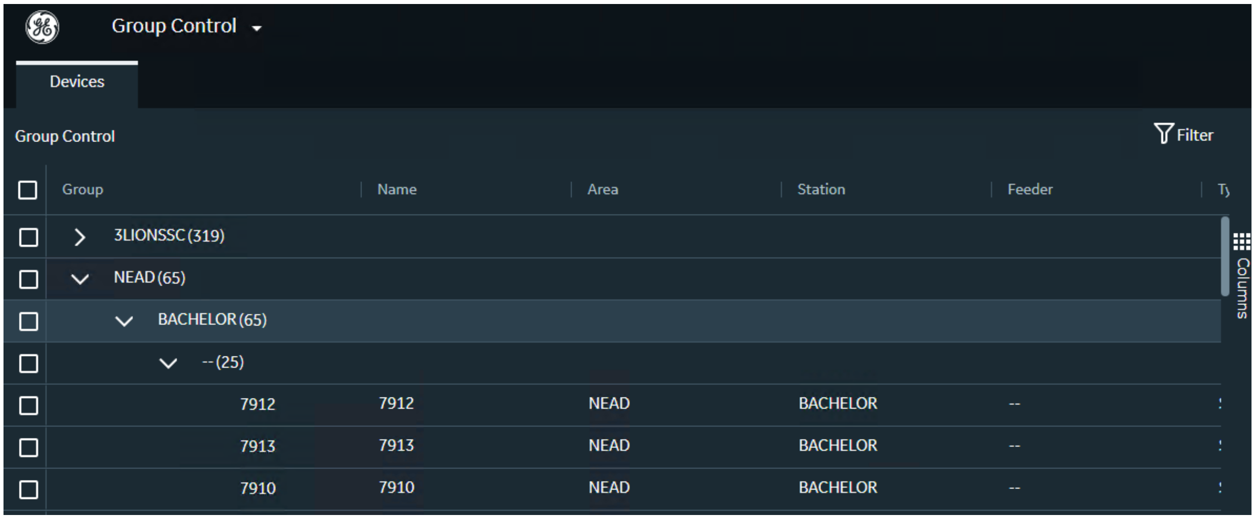


Figure 4. Viewing Components in the Group Control Display

The Group Control grid includes the following fields:

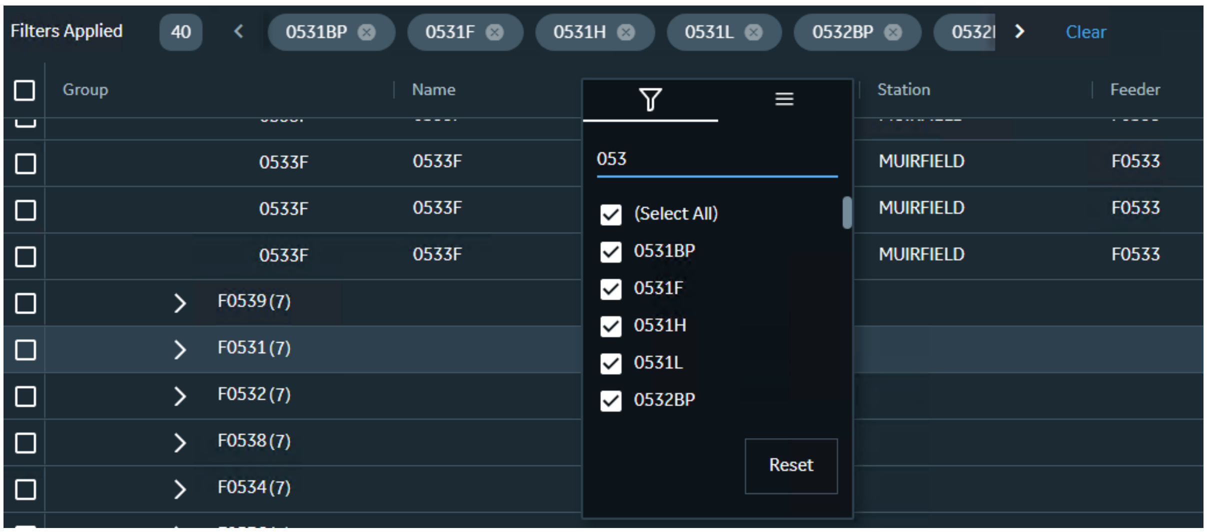
- **Select:** Click to select a component and all lower level components in the grid.
- **ID:** Device ID. This column is hidden by default.
- **Group:** The name of the hierarchical element (service center, station, or feeder).
- **Name:** Device name.
- **Area:** The name of the service center that is associated with the device.
- **Station:** The name of the station that is associated with the device.
- **Feeder:** The name of the feeder that is associated with the device.
- **Type:** The device type.
- **Status:** The operational status of the device.
- **Status Names:** The list of available device statuses. This column is hidden by default.
- **Composite ID Type:** The composite ID of the measurement. This column is hidden by default.
- **Measurement Type:** The type of the measurement.

3.4. Filtering Data on the Display

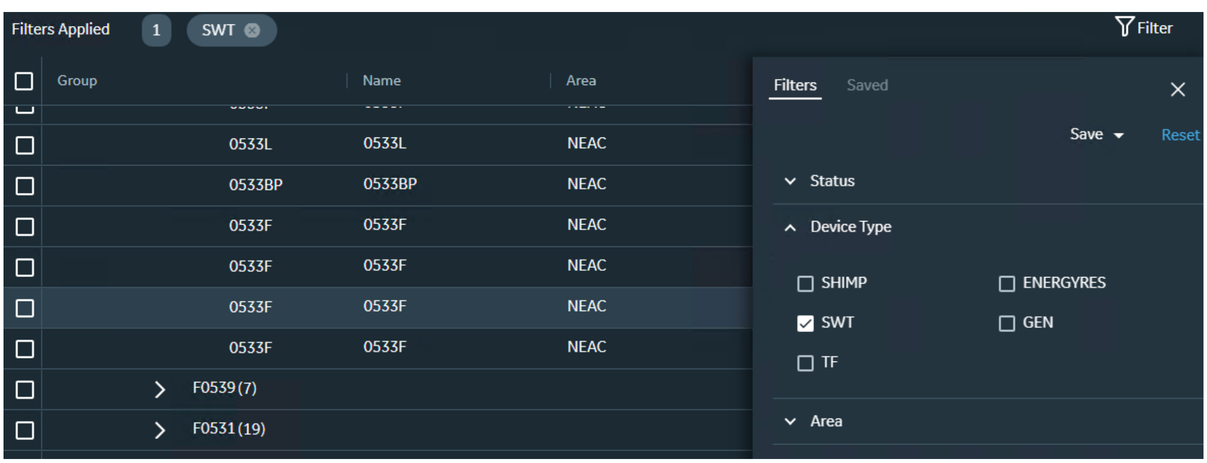
You can filter data based on the specified filtering criteria so that the table is filtered to show only those components that contain the filter text or numbers somewhere in any of the displayed fields. In this way, the results can be filtered by part of a device name, station name, or ID code. Filtering can be removed by clearing the filter text entry field.

The following filtering options are available:

- **Per-Column Filter:** Clicking the column header shows the column filtering button for filtering the display by data in that column.

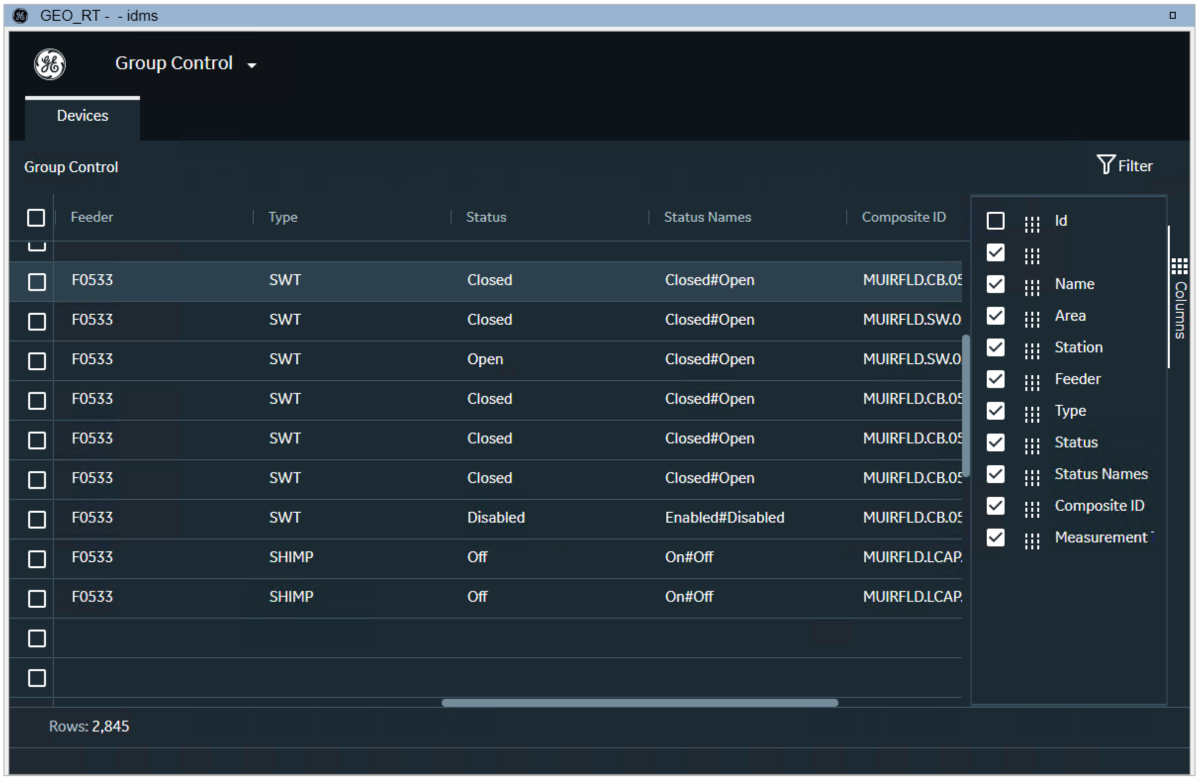


- **Filtering Panel:** Clicking the Show/Hide Filter Panel button Filter shows or hides the filtering panel, where you can customize filtering by columns and save filtering configurations. The Saved tab includes controls to select one of the saved filtering configurations.



- **Filter Criteria:** Shows the current filtering criteria. Clicking a filtering criterion clears it (deselects the associated check box in the filtering box). Clicking the Clear button clears all filtering criteria.

- **Column Filter:** Clicking the Columns button at the right of the display shows the column filtering panel, where you can show or hide columns.



3.5. Selecting Components on the Display

You can select different components from an area, substation, or feeder. To select one or more components, use the selection boxes at the left of the component name in the hierarchy tree. The selection boxes appear for each hierarchy level. You can select components in any combination.

> **Note**
>
> The applied filters are considered when making a selection.

For example, click the section box at the left of a feeder node to select all devices under the feeder. If all child items are selected, the parent node shows the Full Selection icon.

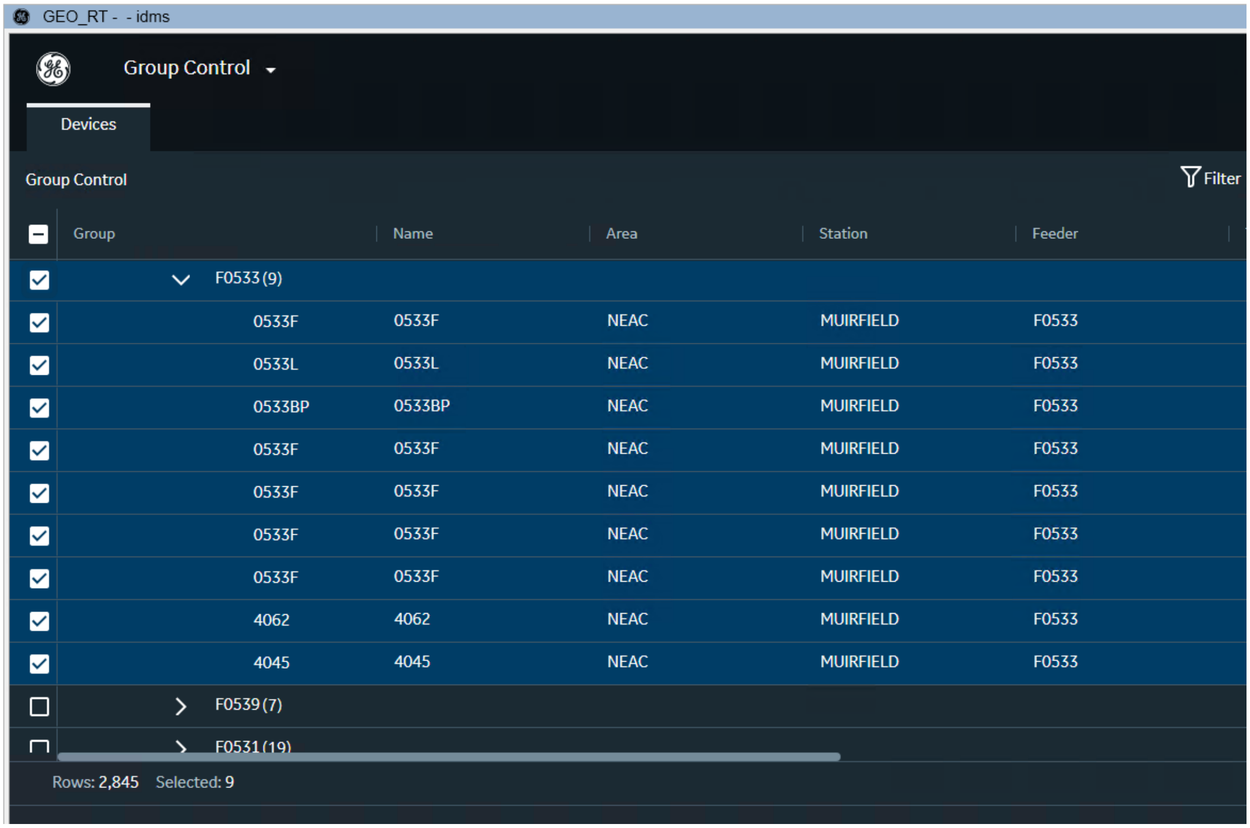


Figure 5. Selecting Components for a Feeder in the Grid

You can refine the selection by deselecting items from the initial group selection. If any items are deselected, the parent node selection will show the Partial Selection icon .

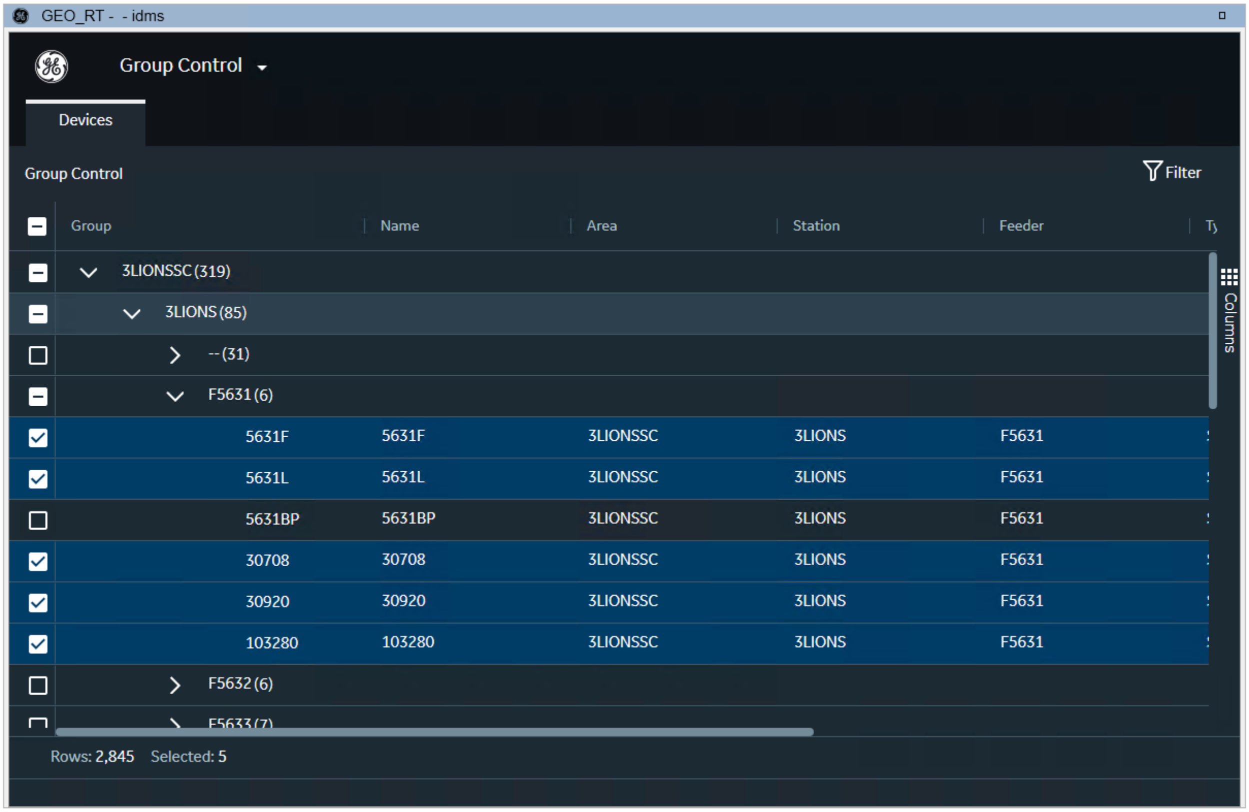


Figure 6. Selecting Mixed Components in the Grid

The status field at the bottom of the table shows the number of components in the selection.

3.6. Issuing a Control

You can issue a control for the selected device using the context menu. The context menu options include all available (applicable) controls types that apply to the selected device types.

For example, when the selection includes both switching devices and capacitors, the context menu includes the Open, Close, On, and Off options, where Open and Close options are associated with switching devices, and On and Off options are associated with capacitors.

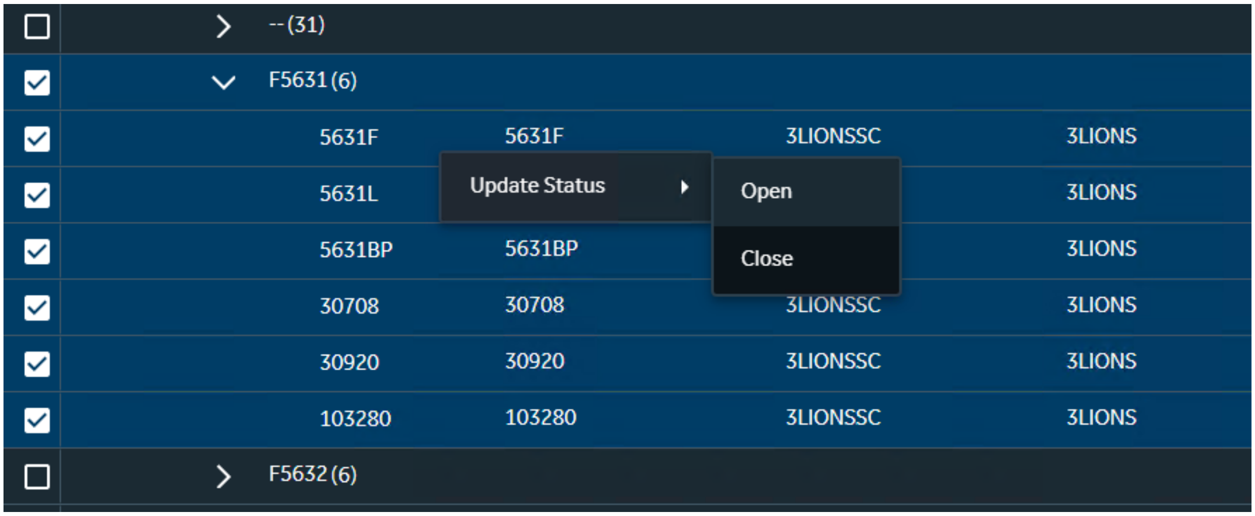


Figure 7. Showing Controls for One Device Type

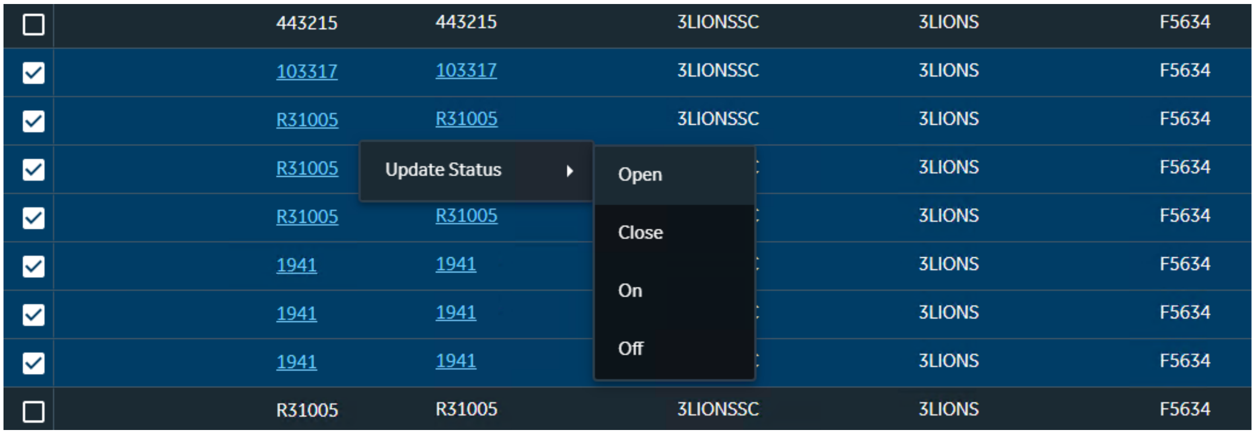


Figure 8. Showing Controls for Multiple Device Types

If more than one device type is selected, the control type is issued to the appropriate device types only. The Control pop-up indicates the device type for which the selected control action will be applied.

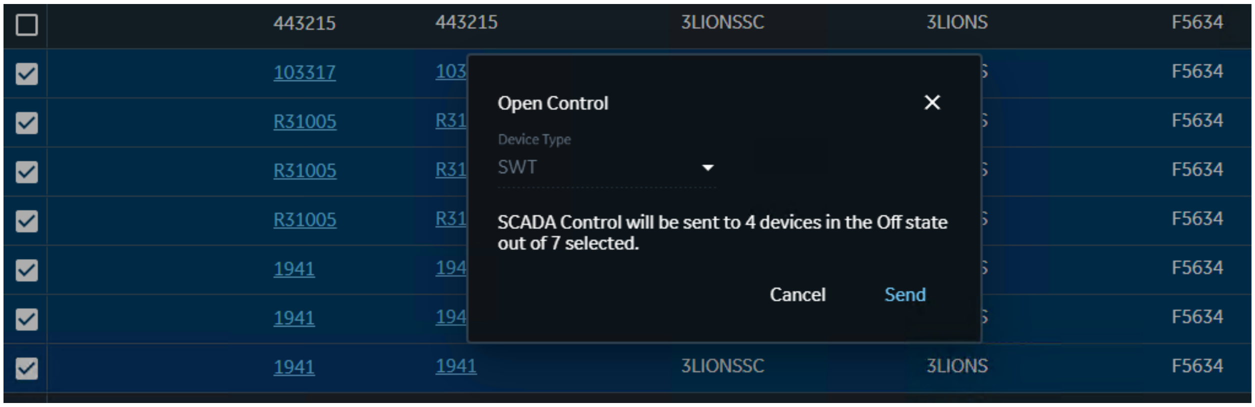


Figure 9. Issuing an Open Control

For example, when you select breakers and capacitors and issue an Open command, the application checks the status of all selected devices and identifies which breakers are at the Closed state and could be opened using the Open command. The Open command is not sent to breakers that are already in the Open state.

When the control is sent, the confirmation notification appears in the application.

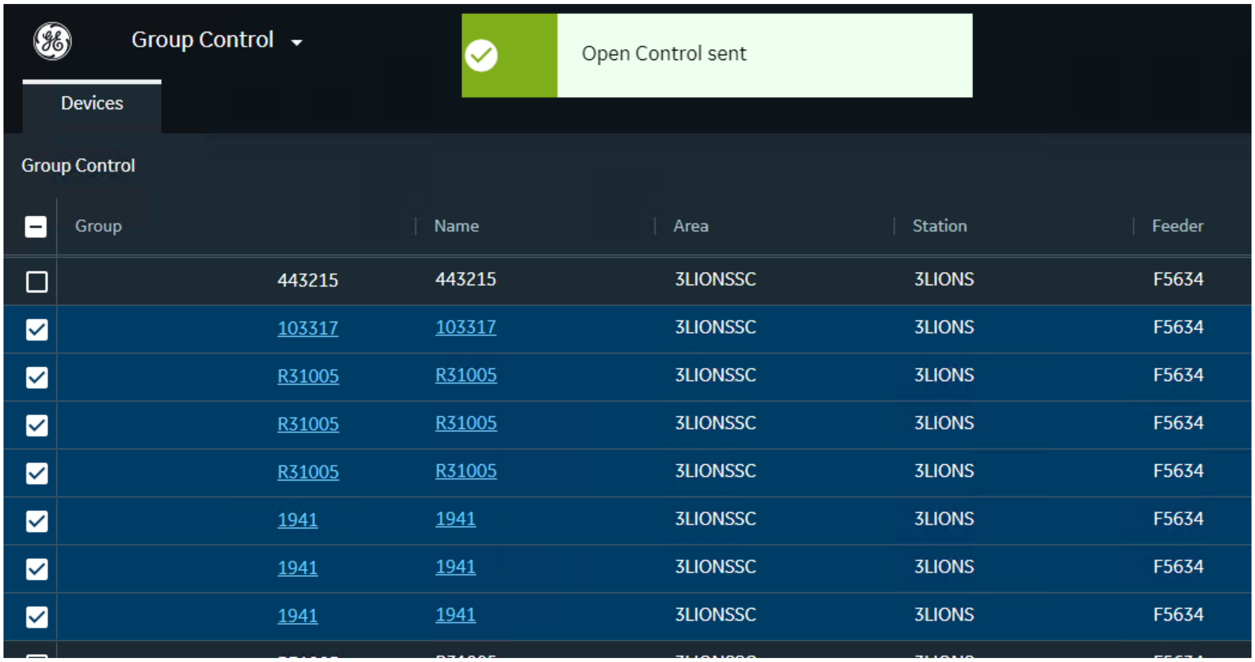


Figure 10. Control Confirmation

When the control is successfully implemented, the device status is updated on the network view. If a control is rejected for any reason in the Scada system, the corresponding entry appears in the Event Log. The control failure is not reported in the Group Control application.

3.7. Supported Group Controls

The Group Control application supports a two-status control point for devices. The following list shows devices that have two status controls, and therefore are supported. For example, to support a new recloser device with two status controls, you need to implement modeling changes only; no code changes are needed.

- Select a group of PCC devices and issue Mode Status control of "Remote"
- Select a group of PCC devices and issue the "Standalone" Mode Status control
- Select a group of PCC devices and issue the "Open" Device Status control
- Select a group of PCC devices and issue the "Close" Device Status control
- Select a group of RAR devices and issue the "Solid" Reclose Status control
- Select a group of RAR devices and issue the "Auto" Reclose Status control

- Select a group of RAR devices and issue the "Open" Device Status control
- Select a group of RAR devices and issue the "Close" Device Status control
- Select a group of RCS devices and issue the "Solid" LVA Status control
- Select a group of RCS devices and issue the "Auto" LVA Status control
- Select a group of RSI devices and issue the "Open" Device Status control
- Select a group of RSI devices and issue the "Close" Device Status control
- Select a group of RSR devices and issue the "Open" Device Status control
- Select a group of RSR devices and issue the "Close" Device Status control
- Select a group of RSR devices and issue the "Solid" Reclose Status control
- Select a group of RSR devices and issue the "Auto" Reclose Status control

4. Demo httpd.conf Configuration

The following httpd.conf file was used for testing on the demo environment.

Note

By default, Group control runs in the HTTPS mode. The demo httpd.conf file references sample self-signed certificates. These certificates must be copied to the specified Apache HTTP Server folder ({SRVROOT}/conf#). These certificates must also be recorded in the system certificate store.

```
# Check if SRVROOT env variable is available. It will be used if defined in system, must use forward
slashes
# Otherwise, uncomment line below to override system SRVROOT
# Define SRVROOT "D:/Eterra/Distribution/GroupControlServices/Apache"

ServerRoot "${SRVROOT}"

Listen 9990 https

# To enable https, configure Listen as follows:
# Listen 9990 https
# To enable http, configure Listen as follows:
# Listen 9990
# Also uncomment lines starting from VirtualHost till SSLStaplingCache

LoadModule actions_module modules/mod_actions.so
LoadModule allowmethods_module modules/mod_allowmethods.so
LoadModule asis_module modules/mod_asis.so
LoadModule auth_basic_module modules/mod_auth_basic.so
LoadModule authn_core_module modules/mod_authn_core.so
LoadModule authn_file_module modules/mod_authn_file.so
LoadModule authz_core_module modules/mod_authz_core.so
LoadModule authz_groupfile_module modules/mod_authz_groupfile.so
LoadModule authz_host_module modules/mod_authz_host.so
LoadModule authz_user_module modules/mod_authz_user.so
LoadModule autoindex_module modules/mod_autoindex.so
LoadModule cgi_module modules/mod_cgi.so
LoadModule deflate_module modules/mod_deflate.so
LoadModule dir_module modules/mod_dir.so
LoadModule env_module modules/mod_env.so
LoadModule http2_module modules/mod_http2.so
LoadModule headers_module modules/mod_headers.so
LoadModule include_module modules/mod_include.so
LoadModule isapi_module modules/mod_isapi.so
LoadModule lbmethod_byrequests_module modules/mod_lbmethod_byrequests.so
LoadModule log_config_module modules/mod_log_config.so
LoadModule mime_module modules/mod_mime.so
LoadModule negotiation_module modules/mod_negotiation.so
LoadModule proxy_module modules/mod_proxy.so
LoadModule proxy_ajp_module modules/mod_proxy_ajp.so
```

```
LoadModule proxy_balancer_module modules/mod_proxy_balancer.so
LoadModule proxy_connect_module modules/mod_proxy_connect.so
LoadModule proxy_html_module modules/mod_proxy_html.so
LoadModule proxy_http_module modules/mod_proxy_http.so
LoadModule proxy_wstunnel_module modules/mod_proxy_wstunnel.so
LoadModule rewrite_module modules/mod_rewrite.so
LoadModule setenvif_module modules/mod_setenvif.so
LoadModule slotmem_shm_module modules/mod_slotmem_shm.so
LoadModule socache_shmcb_module modules/mod_socache_shmcb.so
LoadModule ssl_module modules/mod_ssl.so
LoadModule xml2enc_module modules/mod_xml2enc.so

# 'Main' server configuration

# Uncomment lines staring from VirtualHost till SSLStaplingCache in order to enable https
# Make sure certificate files are available in specified location
# SSLCertificateFile should be installed as Personal / Local Computer
# SSLCertificateChainFile should be installed as Trusted Root CA / Local Computer

# Comment lines staring from VirtualHost till SSLStaplingCache in order to disable https

<VirtualHost _default_:9990>

     ServerName localhost
     SSLEngine on
     SSLCertificateFile "${SRVROOT}/conf/adms-cnv-server.crt"
     SSLCertificateKeyFile "${SRVROOT}/conf/adms-cnv-server.key"
     SSLCertificateChainFile "${SRVROOT}/conf/adms-root-ca.crt"

     # enable HTTP/2, if available
     Protocols h2 http/1.1

     # HTTP Strict Transport Security (mod_headers is required) (63072000 seconds)
     Header always set Strict-Transport-Security "max-age=63072000"

</VirtualHost>

SSLProtocol             all -SSLv3 -TLSv1 -TLSv1.1

SSLHonorCipherOrder     off
SSLSessionTickets       off

SSLUseStapling On
SSLStaplingCache "shmcb:logs/ssl_stapling(32768)"


# ServerAdmin: Your address, where problems with the server should be
# e-mailed.  This address appears on some server-generated pages, such
# as error documents.  e.g. admin@your-domain.com

ServerAdmin admin@example.com
```

```
# Deny access to the entirety of your server's filesystem. You must
# explicitly permit access to web content directories in other
# <Directory> blocks below.

<Directory />
    AllowOverride none
    Require all denied
</Directory>

# DocumentRoot: The directory out of which you will serve your
# documents. By default, all requests are taken from this directory, but
# symbolic links and aliases may be used to point to other locations.

DocumentRoot "${IDMS_ROOT}/webpages"
<Directory "${IDMS_ROOT}/webpages">
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>


# DirectoryIndex: sets the file that Apache will serve if a directory
# is requested.

<IfModule dir_module>
    DirectoryIndex index.html
</IfModule>


# The following lines prevent .htaccess and .htpasswd files from being
# viewed by Web clients.

<Files ".ht*">
    Require all denied
</Files>


# ErrorLog: The location of the error log file.
# If you do not specify an ErrorLog directive within a <VirtualHost>
# container, error messages relating to that virtual host will be
# logged here.  If you *do* define an error logfile for a <VirtualHost>
# container, that host's errors will be logged there and not here.

ErrorLog "logs/error.log"


# LogLevel: Control the number of messages logged to the error_log.
# Possible values include: debug, info, notice, warn, error, crit,
# alert, emerg.

LogLevel warn

<IfModule log_config_module>
```

```
    # The following directives define some format nicknames for use with
    # a CustomLog directive (see below).

    LogFormat "%h %l %u %t \"%r\" %>s %b \"%{Referer}i\" \"%{User-Agent}i\"" combined
    LogFormat "%h %l %u %t \"%r\" %>s %b" common

    <IfModule logio_module>
      # You need to enable mod_logio.c to use %I and %O
      LogFormat "%h %l %u %t \"%r\" %>s %b \"%{Referer}i\" \"%{User-Agent}i\" %I %O" combinedio
    </IfModule>


    # The location and format of the access logfile (Common Logfile Format).
    # If you do not define any access logfiles within a <VirtualHost>
    # container, they will be logged here.  Contrariwise, if you *do*
    # define per-<VirtualHost> access logfiles, transactions will be
    # logged therein and *not* in this file.

    CustomLog "logs/access.log" common

</IfModule>

<IfModule headers_module>

    # Avoid passing HTTP_PROXY environment to CGI's on this or any proxied
    # backend servers which have lingering "httpoxy" defects.
    # 'Proxy' request header is undefined by the IETF, not listed by IANA

    RequestHeader unset Proxy early
</IfModule>

<IfModule mime_module>

    # TypesConfig points to the file containing the list of mappings from
    # filename extension to MIME-type.

    TypesConfig conf/mime.types

    AddType application/x-compress .Z
    AddType application/x-gzip .gz .tgz

</IfModule>



# Customizable error responses come in three flavors:
# 1) plain text 2) local redirects 3) external redirects
#
# Some examples:
#ErrorDocument 500 "The server made a boo boo."
#ErrorDocument 404 /missing.html
#ErrorDocument 404 "/cgi-bin/missing_handler.pl"
```

```
#ErrorDocument 402 http://www.example.com/subscription_info.html

# Secure (SSL/TLS) connections

<IfModule ssl_module>
    SSLRandomSeed startup builtin
    SSLRandomSeed connect builtin
</IfModule>

SSLProxyEngine on

# Tile Cache Service Load Balancing Configuration example
# Uncomment and add more BalancerMember lines, specifying more tile services to load balance

<Proxy balancer://tileconfigset>
    BalancerMember "http://localhost:9991/api/config" loadfactor=1
#   BalancerMember "http://localhost:19991/api/config" loadfactor=1
    ProxySet lbmethod=byrequests
</Proxy>

<Proxy balancer://tileset>
    BalancerMember "http://localhost:9991/api/tile" loadfactor=1
#   BalancerMember "http://localhost:19991/api/tile" loadfactor=1
    ProxySet lbmethod=byrequests
</Proxy>


ProxyPass "/netview/api/config" "balancer://tileconfigset/"
ProxyPassReverse "/netview/api/config" "balancer://tileconfigset/"

ProxyPass "/netview/api/tile" "balancer://tileset/"
ProxyPassReverse "/netview/api/tile" "balancer://tileset/"

# Tile Cache Service Proxy
ProxyPass "/tilecacheservice" "ws://localhost:9991/tilecacheservice"
ProxyPassReverse "/tilecacheservice" "ws://localhost:9991/tilecacheservice"

# Notification Service Proxy
ProxyPass "/notifications" "ws://localhost:9992/notifications"
ProxyPassReverse "/notifications" "ws://localhost:9992/notifications"
ProxyPass "/notify" "http://localhost:9992/notify"
ProxyPassReverse "/notify" "http://localhost:9992/notify"

# Preference Service Proxy
ProxyPass "/preferences" "http://localhost:9994/preferences"
ProxyPassReverse "/preferences" "http://localhost:9994/preferences"

# Assets Proxy
ProxyPass "/grafana/assets" "https://localhost:9990/app-md/assets"
ProxyPassReverse "/grafana/assets" "https://localhost:9990/app-md/assets"

# Grafana Proxy
ProxyPass "/grafana" "http://localhost:9995"
```

```
ProxyPassReverse "/grafana" "http://localhost:9995"

# Netview Adapter Proxy
ProxyPass "/netview/api" "http://localhost:24468/api"
ProxyPassReverse "/netview/api" "http://localhost:24468/api"

# Trending Service Proxy
ProxyPass "/trending" "http://localhost:9996/trending"
ProxyPassReverse "/trending" "http://localhost:9996/trending"

# Graphql Service Proxy
ProxyPass "/v2/graphql" "http://localhost:9997/v2/graphql"
ProxyPassReverse "/v2/graphql" "http://localhost:9997/v2/graphql"

# DMS Interface Server Proxy
ProxyPass "/dms" "http://localhost:8801/dms"
ProxyPassReverse "/dms" "http://localhost:8801/dms"

# Web Maps Load Balancing Configuration example
# Use http://localhost:9990/dark_nolabels/{z}/{x}/{y}.png as Web Map URL

<Proxy balancer://mapset1>
    BalancerMember "http://1.basemaps.cartocdn.com/dark_nolabels" loadfactor=1
    BalancerMember "http://2.basemaps.cartocdn.com/dark_nolabels" loadfactor=1
    BalancerMember "http://3.basemaps.cartocdn.com/dark_nolabels" loadfactor=1
    BalancerMember "http://4.basemaps.cartocdn.com/dark_nolabels" loadfactor=1
    ProxySet lbmethod=byrequests
</Proxy>

# Web Map Proxy 1
ProxyPass "/dark_nolabels" "balancer://mapset1/"
ProxyPassReverse "/dark_nolabels" "balancer://mapset1/"

# Web Maps Load Balancing Configuration example
# Use http://localhost:9990/spotify_dark/{z}/{x}/{y}.png as Web Map URL

<Proxy balancer://mapset2>
    BalancerMember "http://1.basemaps.cartocdn.com/spotify_dark" loadfactor=1
    BalancerMember "http://2.basemaps.cartocdn.com/spotify_dark" loadfactor=1
    BalancerMember "http://3.basemaps.cartocdn.com/spotify_dark" loadfactor=1
    BalancerMember "http://4.basemaps.cartocdn.com/spotify_dark" loadfactor=1
    ProxySet lbmethod=byrequests
</Proxy>

# Web Map Proxy 2
ProxyPass "/spotify_dark" "balancer://mapset2/"
ProxyPassReverse "/spotify_dark" "balancer://mapset2/"
```